

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of the Rising Waters project is to develop a Machine Learning-based flood prediction system that analyzes weather and rainfall parameters to predict the possibility of flooding. The system aims to provide quick, reliable, and accurate predictions to help users take preventive actions before flood events occur.
Scope	The project focuses on predicting flood risk using historical weather and rainfall data. It includes data preprocessing, training a Random Forest machine learning model, developing a Flask-based web application, and deploying the application online to provide instant flood predictions through an easy-to-use interface.
Problem Statement	
Description	Floods cause significant damage to lives, agriculture, and infrastructure every year. Traditional warning methods are often delayed or rely heavily on manual monitoring. There is a need for an intelligent system capable of predicting flood risk quickly using weather and rainfall data.
Impact	The proposed solution enables early flood prediction, helping individuals, farmers, and disaster management authorities take precautionary measures. This reduces potential damage, improves disaster preparedness, and supports faster decision-making.
Proposed Solution	
Approach	The solution uses a Random Forest Machine Learning algorithm trained on historical flood and weather data. The Flask web application accepts user inputs such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and

	sub-rainfall values. The trained model processes these inputs and predicts whether there is a Flood Chance or No Flood Chance .
Key Features	• Machine Learning-based flood prediction using Random Forest. • User-friendly web interface developed using HTML, CSS, and Flask. • Instant flood prediction based on weather parameters. • Real-time prediction through a deployed web application. • Responsive and easy-to-use interface. • Online deployment using Render.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv